

CHAPTER 2: Complex numbers quaternions and geometry

Two number systems.

complex numbers $\mathbb{C} = \{a+bi; a, b \in \mathbb{R}\}$

quaternion $\mathbb{H}$ (Hamilton) = $\{a+bi+cj+dk; a, b, c, d \in \mathbb{R}\}$,

Note that, they all have multiplicative identity 1. and.

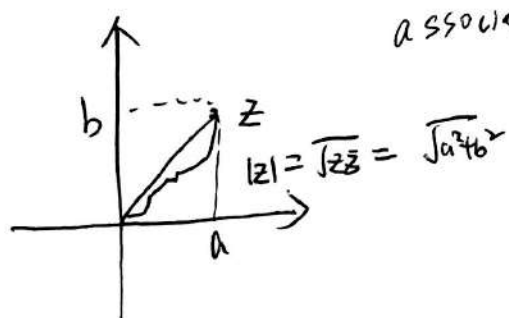
$$|z|^2 = \bar{z}z, |z_1 z_2| = |z_1| |z_2|.$$

Do we have other number system ??? like eight dimensional, or sixteen dimensional?

Let start this chapter by putting these in our mind.

preparation: $\mathbb{C} = \{a+bi; a, b \in \mathbb{R}\}$. $z = a+bi$ $\bar{z} = a-bi$

(associative, commutative)



$$\text{If } |z|=1 \Rightarrow z = e^{i\alpha} = \cos \alpha + i \sin \alpha.$$

like Kuramoto model.

$$\bar{z} = \frac{1}{N} \sum_{i=1}^N e^{i\theta_i} = \underbrace{r}_\text{order parameter} e^{i\alpha}$$

We can extend complex conjugation. to matrices and vectors

$$A = (a_{ij}) \quad (a_{ij} \in \mathbb{C}) \quad \bar{v} = (\bar{v}_1, \bar{v}_2, \dots, \bar{v}_n) \quad (\bar{v}_i \in \mathbb{C})$$

Two important group in chapter 6:

$$1) \text{ Unitary group } U(n, \mathbb{C}) = \{M \in M(n, \mathbb{C}); \bar{M}^T M = I_n\}$$

$$M \in U(n, \mathbb{C}) \Rightarrow \det(M) = \pm 1.$$

$$2) \text{ Special unitary group } SU(n, \mathbb{C}) = \{M \in U(n, \mathbb{C}); \det(M) = 1\}$$

• quaternions. $\mathbb{H} = \{a+bi+cj+dk, a, b, c, d \in \mathbb{R}\}$

(Why??) $i^2 = j^2 = k^2 = -1$ $ij = k = -ji$ $jk = i = -kj$ $ki = j = -ik$.

Since Hamilton want to extend complex number to quaternions.

$$i^2 = j^2 = k^2 = ijk = -1 \Rightarrow ij = k = -ji, \text{ (associative, non-commutative)}$$

addition: $z = a+bi+cj+dk$ $z' = a'+b'i+c'j+d'k$ $z+z' = a+a' + (b+b')i + (c+c')j + (d+d')k$ ②

multiplication: $zz' = (aa' - bb' - cc' - dd') + (ab' + ba' - dc')i + (ac' + ca' + da' - ad')j + (ad' + da' + bc - cb')k$

What is identity for addition and multiplication? $0 = (1, 0, 0, 0)$

$(xz)y = x(zy)$

associative

$x(z+y) = xz + xy$

distributive law

$\bar{z} = a-bi-cj-dk$

$|z| = \sqrt{\bar{z}z} = \sqrt{z\bar{z}} \quad ???$
 $= a^2 + b^2 + c^2 + d^2$

Main Results:

T2.1. Assume T is in $O(\mathbb{R}^3)$ and T has matrix M (standard basis) Then T has determinant 1 $\Leftrightarrow T$ is a rotation about the line through the origin.

If line is the z -axis, we have

$$M = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

proof: $O(\mathbb{R}^3) = \{ T \in GL(\mathbb{R}^3) : T \text{ preserves distance} \}$

$GL(\mathbb{R}^3) = \{ T: \mathbb{R}^n \rightarrow \mathbb{R}^n : T \text{ linear and invertible} \}$

We prove it into two steps.

Step A: T is a rotation $\Rightarrow \det(M) > 0$.

$T \in O(\mathbb{R}^3) = \{ M \in M(3, \mathbb{R}) : M^T M = I_3 \} \Rightarrow \det(MM^T) = \det(M)\det(M^T)$
 $= (\det(M))^2 = \det(I_3) = 1$

$\Rightarrow \det(M) = \pm 1$ by T is a rotation $\Rightarrow \det(T) = 1$

$T = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Step B: $\det(M)=1 \Rightarrow T$ is a rotation.

③ 3

$$\det(M)=1 \Rightarrow \det(I_3 - M) = \det(M^T M - M) = \det(M^T - I_3) \det(M) \\ = \det(M^T - I_3) \det(M) \stackrel{=1}{=} \det(M^T - I_3) = \det(M - I_3) = (-1)^3 \det(I_3 - M)$$

$$\Rightarrow \det(I_3 - M) = 0 \Rightarrow \exists a \neq 0 \text{ s.t. } (M - I_3)a = 0$$

$\Rightarrow a$ is the eigenvector of eigenvalue 1.

Choose space $U = \{u \in \mathbb{R}^3 : u \cdot a = 0\}$ $\dim(U) = 2$. a is the axis.

Next, we show that T is the rotation with axis a .

Step B.1 By $T \in O(\mathbb{R}^3) \Rightarrow \|T(a) - T(b)\| = \|a - b\|$.

T preserve distance $\Rightarrow T$ is rotation.

$\forall x \in \mathbb{R}^3 \quad x = \alpha a + \beta e_1 + \gamma e_2$. (e_1, e_2 is the basis of U)

$$T(x) = \alpha a + \beta T(e_1) + \gamma T(e_2).$$

Step B.1. $T(U) = U$.

$T \in O(\mathbb{R}^3) \Rightarrow T$ preserve linear product.

Case B.1.1 $T(U) \subset U$. $\forall x \in T(U)$. $x \cdot a = T(v) \cdot T(a) = v \cdot a = 0$
 $\Rightarrow x \cdot a = 0 \Rightarrow x \in U$.

Case B.1.2. $U \subset T(U)$. We only need to show $\dim(T(U)) = 2$.

$$T(e_1) \cdot T(e_2) = e_1 \cdot e_2 = 0 \quad T(\alpha e_1 + \beta e_2) = \alpha T(e_1) + \beta T(e_2)$$

$$\forall y \in T(U), \exists x \in U = \alpha e_1 + \beta e_2 \quad y = \alpha T(e_1) + \beta T(e_2).$$

The proof is completed.

If we take $a = (0, 0, 1)$ $U = xy$ axis.

$T: U \rightarrow U \Leftrightarrow T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear preserving distance.

by page 23 $\Rightarrow T = \left\{ \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}, \begin{bmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & -\cos \alpha \end{bmatrix} \mid \alpha \in \mathbb{R} \right\}$ (4)

by $\det(M) = 1 \Rightarrow T' = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$

$\Rightarrow M = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$

T2.2. (a) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$, $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$, $|z_1 z_2| = |z_1| |z_2|$ $\square$. easy

(b) $z_1 z_2 = z_2 z_1$ (Abelian)

group: ① closure ② associative $(a+b)+c = a+(b+c)$

③ Identity ④ Inverse element.

In summary: $(\mathbb{C}, +)$ is an abelian group.

(c) $(\mathbb{C} \setminus \{0\}, \times)$ is an abelian group.

$(a+bi)(a'+b'i) = (aa' - bb') + (ab' + ba')i$

① closure ② associative ③ Identity $(1,0)$.

④ Inverse element. $(a+bi)(a'+b'i) = (1,0) \Rightarrow a' = \frac{a}{a^2+b^2}$ $b' = -\frac{b}{a^2+b^2}$

$\begin{cases} ab' + ba' = 0 \\ aa' - bb' = 1 \end{cases} \leftarrow \begin{aligned} a' &= \frac{1+bb'}{a} \quad ab' + \frac{b}{a}(1+bb') = 0 \Rightarrow b' \left(\frac{b^2}{a} + a \right) = -\frac{b}{a} \\ \Rightarrow b' &= \frac{-\frac{b}{a}}{\frac{b^2+a^2}{a}} = -\frac{b}{a^2+b^2} \end{aligned}$

$(S', \times)$ is a subgroup of $(\mathbb{C}, \times)$.

① S' is closed ② S contains $(1,0)$ ③ Inverse $a^2+b^2 \neq 0 \Rightarrow a' = a$ $b' = -b$.

④ $|z_1| |z_2| = |z_1 z_2|$

$\square$.

T2.3. $U(n, \mathbb{C})$ is a group under matrix multiplication, and $SU(n, \mathbb{C})$ is a subgroup. In the special case $n=2$, every matrix in $SU(2, \mathbb{C})$ has the form

$$M = \begin{bmatrix} u & -\bar{v} \\ v & \bar{u} \end{bmatrix} \text{ for } u, v \text{ satisfying } u\bar{u} + v\bar{v} = 1$$

Proof: (*) $U = \{ M \in M(n, \mathbb{C}) : M^T M = I \}$ is a group under $\times$

① closed. $M_1^T M_1 = I \quad M_2^T M_2 = I \quad (M_1 M_2)^T M_1 M_2 = M_2^T M_1^T M_1 M_2 = I$

② associative $M_1 (M_2 M_3) = (M_1 M_2) M_3$

③ identity $I_n = \begin{bmatrix} (1,0) & & \\ & (1,0) & \\ & & \ddots & \\ & & & (1,0) \end{bmatrix}$

④ Inverse $M^T M = I_n \Rightarrow M^T = M^{-1}$ why??

$$M^T = M^T M C = I C = C \Rightarrow M^T = C \Rightarrow M^T = M^{-1}$$

(*) $SU(n, \mathbb{C}) = \{ M \in U(n, \mathbb{C}) : \det(M) = 1 \}$

① closed $\det(M_1 M_2) = \det M_1 \cdot \det M_2 = 1$

② associative $\checkmark$

③ identity $I_n \in SU(n, \mathbb{C}) \quad \det(M^T) = \det(M) = 1$

④ Inverse $M^T \in SU(n, \mathbb{C})$ if $M \in SU(n, \mathbb{C})$

$$(*) \quad n=2 \Rightarrow M = \begin{bmatrix} u & -\bar{v} \\ v & \bar{u} \end{bmatrix}$$

We set $M = \begin{bmatrix} u & w \\ v & z \end{bmatrix} \quad \det(M) = 1 \Rightarrow uz - vw = 1 \quad (*)$

$$M^T M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \bar{u} & \bar{w} \\ \bar{v} & \bar{z} \end{bmatrix}^T \begin{bmatrix} u & w \\ v & z \end{bmatrix} = \begin{bmatrix} \bar{u} & \bar{v} \\ \bar{w} & \bar{z} \end{bmatrix} \begin{bmatrix} u & w \\ v & z \end{bmatrix} = \begin{bmatrix} \bar{u}u + \bar{v}v & \bar{u}w + \bar{v}z \\ \bar{w}u + \bar{z}v & \bar{w}w + \bar{z}z \end{bmatrix}$$

$$\Rightarrow \bar{u}u + \bar{v}v = 1 \quad \bar{w}w + \bar{z}z = 1 \quad \bar{v}\bar{v} = 1 \Rightarrow \bar{v} = \bar{v} \quad \text{why??}$$

Case A. $u=0$ by (*) $\Rightarrow v w = -1 \Rightarrow w = -v^{-1} = -\bar{v}$

Since $\bar{u}w + \bar{v}z = 0$ and $\bar{w}u + \bar{z}v = 0$ $\Rightarrow w = -\frac{\bar{v}z}{\bar{u}}$ $\Leftarrow$ Case B $u \neq 0$

$$w = -\frac{\bar{v}z}{u} \quad wz = \frac{1+uz}{v} \quad z = \frac{-}{u} \quad \text{why } z\bar{z} \Rightarrow w = -\bar{v} \text{ by (1)}$$

$$\Rightarrow \bar{u} - \bar{u}uz = v\bar{v}z \Rightarrow (\bar{u}u + v\bar{v})z = \bar{u}$$

2.4. Quaternions conjugation.

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2 \quad \overline{z_1 z_2} = \bar{z}_2 \bar{z}_1$$

We can set $z_1 = a_1 + w_1 \quad z_2 = a_2 + w_2$

$$w_1 = (c_1, j_1, k) \quad w_2 = (c_2, j_2, k)$$

$$\bar{z}_1 = a_1 - w_1 \quad \bar{z}_2 = a_2 - w_2$$

$$\Rightarrow \overline{z_1 z_2} = a_1 a_2 - a_1 w_2 - a_2 w_1 - w_1 w_2 = \bar{z}_2 \bar{z}_1$$

$$a_1 a_2 + a_1 w_1 + a_2 w_2 + w_1 w_2 = a_1 a_2 - w_1 w_2 - a_2 w_1 - a_1 w_2$$

(B) The set $(\mathbb{H}, +)$ is an abelian group.

1. closed 2. identity 0. 3. associative

4. Inverse $(-a, -b, -c, -d)$ 5. $z_1 + z_2 = z_2 + z_1$ (commutative)

(C) $(\mathbb{H} \setminus \{0\}, \times)$ is a non-abelian group.

1. closed $z \bar{z} = a^2 + b^2 + c^2 + d^2$

2. Identity $I = (1, 0, 0, 0)$ 3. associative $(z_1 z_2) z_3 = z_1 (z_2 z_3)$

4. Inverse $z^{-1} = \frac{\bar{z}}{a^2 + b^2 + c^2 + d^2} = I$

5. non-abelian $(0, 0, 1, 0) (0, 1, 0, 0) = (0, 0, 0, -1)$
 $(0, 1, 0, 0) (0, 0, 1, 0) = (0, 0, 0, 1)$

(D) The 3-sphere S^3 is a non-abelian subgroup:
of $(\mathbb{H} \setminus \{0\}, \times)$

1. closed $|z_1|=1$ $|z_2|=1$ $(z_1, z_2 | z_1, z_2) = 1$
 $= \sqrt{z_1, z_2 \bar{z}_1 \bar{z}_2} = \sqrt{z_1 \bar{z}_1 z_2 \bar{z}_2} = 1$

2. identity $(1, 0, 0, 0) \in S^3$

3. $|z|=1$ $|\bar{z}|=1$ $z^{-1} = \bar{z}$ (Inverse)

4. $(z_1, z_2) z_3 = z_1 (z_2 z_3)$ associative.

T2.5. (A). Every quaternion q in S^3 can be written as

(why??) $q = \cos(\frac{\alpha}{2}) + \sin(\frac{\alpha}{2}) u$ $u \in S^3$ (Real(u)=0)

$a^2 + b^2 + c^2 + d^2 = 1 \Rightarrow |q| = \cos^2(\frac{\alpha}{2}) + \sin^2(\frac{\alpha}{2}) = 1$

$\bar{q} = \cos \frac{\alpha}{2} - \sin \frac{\alpha}{2} u$ $q^{-1} = \bar{q}$

(B). For $q \in S^3$, $v \in \mathbb{R}^3$. $T_q(v) = q v q^{-1}$ defines a rotation in $SO(\mathbb{R}^3)$. If $q = \cos \frac{\alpha}{2} + \sin \frac{\alpha}{2} u \Rightarrow T_q$ is a rotation through angle α around an axis spanned by u .

1. well-defined $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ $\bar{q} v q^{-1} = \bar{q}^{-1} \bar{v} \bar{q} = q(-v) q^{-1} = -q v q^{-1}$
 $\Rightarrow \text{Re}(T(v)) = 0 \Rightarrow T_q(v) \in \mathbb{R}^3$

2. $|T_q(v)| = |v|$ since $|q v q^{-1}| = |q| |v| |q^{-1}| = |v| \Rightarrow T_q \in O^3$.
 $T_1 = I \Rightarrow \det(T_1) = 1$ $\det(T_q) = \det(q, \cdot) = 1$ $\leftarrow$ T_q is continuous, $\det$ is continuous

3. T is linear $T_q(v+w) = T_q(v) + T_q(w)$ $T_q(c v) = c T_q(v)$

4. $\det(T) = 1 \Rightarrow T$ is a rotation.

Actually if $q = \cos \frac{\alpha}{2} + \sin \frac{\alpha}{2} u$ T_q ~~is~~ (u, v, w) is basis of U
 ~~$T(u, v, w)$~~ $T(u, v, w) = \begin{pmatrix} \cos \frac{\alpha}{2} & -\sin \frac{\alpha}{2} & 0 \\ \sin \frac{\alpha}{2} & \cos \frac{\alpha}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Denote $q = \cos \frac{\theta}{2} + \sin \frac{\theta}{2} u = c + su$ $u \in S^3$ $u^2 = -1$. (8)

$$q^{-1} = \cos \frac{\theta}{2} - \sin \frac{\theta}{2} u =: c - su.$$

For any $x \in \mathbb{R}^3$.

Note that $\vec{u} \vec{v} = -(\vec{u} \cdot \vec{v}) + \vec{u} \times \vec{v}$

$$T_q(x) = \frac{qxq^{-1}}{(\star)} \quad (\text{Rodrigues formula})$$

$$uv = (u_1, u_2, u_3)(v_1, v_2, v_3) = (-u_1v_1 - u_2v_2 - u_3v_3) + (u_2v_3 - u_3v_2)i + (u_3v_1 - u_1v_3)j + (u_1v_2 - u_2v_1)k = -(u \cdot v) + \vec{u} \times \vec{v}$$

By $(\star)$ $T_q(x) = cxc - cxsu + suxc - s^2uxu.$

$$= c^2x - csxu + scux - s^2uxu.$$

$$=: c^2x + I_1 + I_2 + I_3$$

$$I_1 = -csxu = cs(x \cdot u - \vec{x} \times \vec{u}). \quad \because a \times b = (b \times a) \cdot H$$

$$\Rightarrow I_1 + I_2 = -2cs(x \cdot u)$$

$$I_2 = scux = cs(-u \cdot x + \vec{u} \times \vec{x})$$

$$I_3 = -s^2uxu = -s^2(-u \cdot x + \vec{u} \times \vec{x}) \vec{u}$$

$$= s^2(u \cdot x) \vec{u} - s^2((\vec{u} \times \vec{x}) \cdot \vec{u})$$

$$= s^2(u \cdot x) \vec{u} - s^2(-(\vec{u} \times \vec{x}) \cdot \vec{u} + \vec{u} \times \vec{x} \times \vec{u})$$

$$= 0$$

$$= s^2(u \cdot x) \vec{u} - s^2(x \cdot (u \cdot u) - u(x \cdot u))$$

$$u \cdot u = 1$$

$$= -s^2x + s^2u(x \cdot u).$$

$$\Rightarrow T_q(x) = (c^2 - s^2)x + 2cs(\vec{u} \times \vec{x}) + s^2u(x \cdot u)$$

$$c^2 - s^2 = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \cos \theta \quad 2cs = \sin \theta \quad 2s^2 = 1 - \cos \theta$$

$$\Rightarrow T_q(x) = x \cos \theta + (\vec{u} \times \vec{x}) \sin \theta + u(u \cdot x)(1 - \cos \theta).$$

$$T_q(u) = u \quad \text{If } u = vxw, \quad w, v \text{ are the basis.}$$

$$T_\theta(v) = v \cos \theta + (u \times v) \sin \theta = v \cos \theta - w \sin \theta \quad (9)$$

$$T_\theta(w) = w \cos \theta + (u \times w) \sin \theta = v \sin \theta + w \cos \theta$$

$$\Rightarrow M = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow T \text{ is a rotation by } T_{2,1}$$

(C) $F: S^3 \rightarrow \text{SO}(R^3)$ is onto but not one-to-one.
 $(u \mapsto -u, \alpha \mapsto \alpha + \pi)$

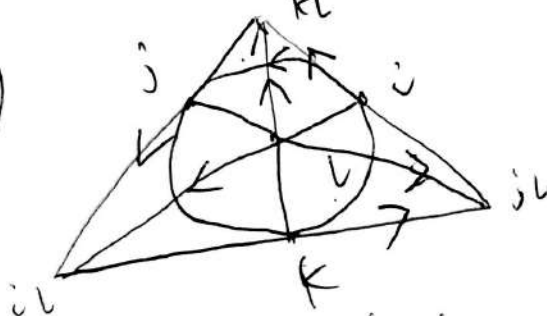
proof: $F(q) = q \cup q^{-1} \quad \forall v \in \text{SO}(R^3), \exists q \text{ s.t. } F(q) = T_q = v$

by T2.1 $T \in \text{SO}(R^3) \Rightarrow T$ is a rotation about the line through the origin. Take α be the rotation angle and v be the unit vector for the line $\Rightarrow q = \cos \frac{\alpha}{2} + \sin \frac{\alpha}{2} u$ (onto) $\mathbb{R}$

2.5 Octonions.

$$\mathbb{O} = \{a + bi + cj + dk + el + fkl + gij + hli, a, b, c, d, e, f, g, h \in \mathbb{R}\}$$

String theory



$$(kl)j = il$$

$$j(kl) = -il$$

Question: Can you create higher and higher dimensional number system? $\mathbb{R} \subset \mathbb{C} \subset \mathbb{H} \subset \mathbb{O}$

If we require identity to be 1 $|z|^2 = z \bar{z}$ ($|z_1 z_2| = |z_1| |z_2|$)
 Then ~~no~~!!! by Hurwitz.